

# **Software Requirements Specification (SRS)**

## **Netflix Content Recommendation System**

### **Project Title:**

Netflix Content Recommendation System

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#### **1. Introduction:**

The purpose of this SRS is to list the different requirements and present a project plan for building a content recommendation system for the brand, which will be used to recommend personalized content to the users, hence improving user engagement. The recommendations will be based on user watch history, watch time, ratings, genres, languages preferred, and overall user behaviour.

This document serves as a bridge between stakeholders and the technical team, which is in charge of developing and integrating this system into the product.

Business benefits include:

- a. Increased customer retention
- b. Increase in watch time
- c. Better recommendation accuracy
- d. Increase in user ratings
- e. Reduced Customer churn

Document Conventions included:

- a. **Requirement identifiers:** Functional requirements are labeled with the prefix FR and non-functional requirements with NFR.
- b. **Priority levels:** Requirements are classified into High, Medium, and Low based on their impact level and business importance.
- c. **Version control:** Any future modifications to this document will be included in the revision history.

Project Scope:

- a. Collecting and processing users' watch history
- b. Storing the platform's metadata
- c. User profile management
- d. Recommending Top-5 movies and shows to users
- e. Allowing users to confirm whether a content suggestion is correct or not.
- f. Generating recommendations using Hybrid content recommendation systems.

Success Criteria:

The project will be considered successful if:

- a. Users' watch time increases
- b. Users' feedback improves
- c. Recommendations appear in acceptable response times.
- d. Users are easily able to discover new content based on their interests.

Intended Audience and Reading Suggestions:

This SRS is intended for all stakeholders involved (and/or) affected by this project. They might be involved in the development, implementation, testing, or maintenance phases.

The intended audience includes:

| Stakeholder              | Purpose                                                                          |
|--------------------------|----------------------------------------------------------------------------------|
| Project Sponsor / Client | To understand the project's objectives, scope, and expected business outcomes.   |
| Business Analysts        | To validate the project's requirements and drive business growth.                |
| Project Manager          | To estimate project timelines, allocate resources, and monitor project progress. |

|                                                   |                                                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Architects</b>                          | To understand the overall system architecture and technical constraints.                                                            |
| <b>Developers</b>                                 | To make amendments to the platform (if required) and integrate the recommendation model into the system via an API.                 |
| <b>Data Scientists and ML Engineers</b>           | To develop and integrate the recommendation model into the system.                                                                  |
| <b>Quality Assurance (QA) Engineers / Testers</b> | To ensure that the platform works properly according to the user/client's expectations after the new changes have been implemented. |
| <b>Data Engineers</b>                             | Building pipelines to ensure high-quality data availability for the recommendation engine.                                          |
| <b>Maintenance Team</b>                           | To understand the system requirements for future enhancements, debugging, and support.                                              |

**Reading Suggestions:**

- a. Project Managers should refer to the Introduction, Project Scope, Overall Description, and System Features to support project planning, resource allocation, and progress monitoring.
- b. Data Scientists / Machine Learning Engineers should focus on the system features to design, develop, and optimize the recommendation algorithm.
- c. Data Engineers should review the Data Requirements, External Interface Requirements, and Database Requirements to understand data ingestion, preprocessing, storage, and pipeline requirements.
- d. Software developers should review all the specified requirements so that they can integrate the recommendation system into the platform using an API.
- e. Quality Assurance (QA) Engineers / Testers should review the product requirements and Success Criteria to design, execute, and confirm all test cases to confirm whether the project has achieved its objectives.

**2. Overall Description:**

The content recommendation model is a hybrid machine-learning model designed to assist users in discovering new content that aligns with their interests. This is made possible by the model by analyzing historical data, content metadata, and user preferences.

The model compares the characteristics of movies and TV shows such as genres, cast and crew, keywords, and descriptions to identify similar content and recommend it to users based on their preferences; hence reducing their time and effort to manually browse through all content available on the platform.

**Product Perspective:**

The Netflix Content Recommendation System is an ML model that can be integrated into the existing platform using an API. This model focuses on generating movie and TV show recommendations for users on the platform. This model utilizes hybrid-based filtering techniques for generating recommendations.

**Product Features:**

- a. Similar Content Recommendation
- b. Content Metadata Analysis
- c. Recommendations Ranking

**User Classes and Characteristics:**

**a. End User:**

These users interact with the recommendation engine to discover new content tailored for them that matches their interests.

**Characteristics:**

- a. Possesses basic computer literacy
- b. Views personalized recommendations generated by the system
- c. Does not have any technical knowledge behind the recommendation engine

**b. Administrator:**

This user is responsible for maintaining the platform dataset and ensuring the recommendation system API fetches results as expected.

**Characteristics:**

- a. Manages content metadata
- b. Checks for any updates to the dataset
- c. Manages users
- d. Monitors system performance and recommendation quality.

**c. Data Scientist:**

These users develop and make improvements to the recommendation model.

**Characteristics:**

- a. Collects and preprocesses the data
- b. Performs feature engineering
- c. Performs exploratory data analysis.
- d. Develops the hybrid (content-based + collaborative-based) recommendation model.
- e. Monitors recommendation accuracy and recommends improvements.

**d. ML Engineer:**

These users make necessary improvements and deploy the ML model developed by them or the data scientists.

**e. Data Engineer:**

The Data Engineer manages the data infrastructure required for the recommendation system.

**Characteristics:**

- a. Builds and maintains data pipelines.
- b. Ensures the availability of clean and reliable movie metadata.
- c. Performs data validation and transformation.
- d. Optimizes data storage and retrieval for the recommendation engine.

**Operating Environment:**

The content recommendation model will operate in a cloud-based environment to provide a recommendation service to all platform users.

**Hardware Environment:**

The system shall operate on cloud-based computing infrastructure capable of supporting large-scale machine learning inference and high user traffic. The infrastructure shall provide sufficient CPU, memory, and storage resources to ensure efficient recommendation generation.

**Operating System:**

The production environment shall use Linux-based operating systems for hosting application services and machine learning models.

**Cloud Environment:**

The recommendation system shall be deployed on Amazon Web Services (AWS).

The cloud environment shall support:

- a. High availability
- b. Automatic scaling
- c. Load balancing
- d. Fault tolerance
- e. Disaster recovery
- f. Secure storage of application data and machine learning artifacts

**Software Environment:**

The production environment shall include:

- a. Python Runtime
- b. Machine Learning inference service
- c. REST API for recommendation requests
- d. Containerized application services (if applicable)

**Database Environment:**

The system shall utilize cloud-hosted databases for storing:

- a. Movie and TV show metadata
- b. User interaction history
- c. User preferences
- d. Recommendation logs

**Network Environment:**

The recommendation service shall communicate securely with other Netflix services through encrypted HTTPS communication within the cloud infrastructure.

**Design and Implementation Constraints:**

The development and deployment of the recommendation model shall follow these design and implementation constraints:

**a. Cloud Infrastructure:**

The recommendation system shall be deployed within the organization's approved cloud infrastructure to ensure scalability, availability, and operational reliability.

**b. Performance:**

The system shall generate personalized recommendations within the response time defined in the non-functional requirements, even during periods of high user traffic.

**c. Scalability:**

The system shall support horizontal scaling to accommodate increasing numbers of users, recommendation requests, and content items without significant degradation in performance.

**d. Security and Privacy:**

The system shall comply with organizational security policies and applicable data privacy regulations. User interaction data shall be securely stored, transmitted, and accessed only by authorized services.

**e. Data Quality:**

The recommendation engine shall rely on accurate, complete, and consistently formatted content metadata and user interaction data. Poor data quality may negatively impact recommendation accuracy.

**f. System Integration:**

The recommendation system shall integrate seamlessly with existing Netflix services through standardized APIs without affecting the performance of other platform components.

**g. Availability:**

The recommendation service shall be designed for high availability and fault tolerance to ensure uninterrupted recommendation delivery.

**h. Maintainability:**

The system architecture shall support model updates, retraining, and software maintenance with minimal disruption to production services.

**i. Monitoring and Logging:**

The system shall maintain logs for recommendation requests, system errors, and operational metrics to facilitate monitoring, troubleshooting, and performance analysis.

**j. Technology Standards:**

The implementation shall comply with Netflix's software development standards, coding guidelines, deployment procedures, and architectural principles.

**Assumptions and Dependencies:**

**Assumptions:**

- a. Users have an active Netflix account and sufficient interaction history to enable personalized recommendations.
- b. Content metadata (such as genre, cast, director, language, description, and release year) is accurate, complete, and regularly maintained.
- c. User viewing history, ratings, and interaction data are continuously collected and made available to the recommendation engine.
- d. The recommendation service operates within Netflix's cloud infrastructure and has access to the required computing resources.
- e. External services and APIs required by the recommendation system remain available and function as expected.
- f. Users access the recommendation system through supported Netflix applications, including web, mobile, and smart TV platforms.
- g. Recommendation requests are received through authenticated internal Netflix services.

**Dependencies:**

- **User Interaction Data:**  
The system depends on user activity data, including:
  - Viewing history
  - Watch duration
  - User ratings
  - Search history
  - Watchlist
- **Content Metadata Repository:**  
The recommendation engine depends on an up-to-date repository containing:
  - Movie and TV show metadata
  - Genres
  - Cast
  - Directors
  - Descriptions
  - Release information
  - Language information
- **Cloud Infrastructure:**  
The recommendation service depends on the availability of the organization's cloud infrastructure for:
  - Model hosting
  - Compute resources
  - Storage
  - Networking
  - Auto-scaling
- **Database Services:**  
The system depends on database services for storing and retrieving:
  - User profiles
  - Viewing history
  - Recommendation logs
  - Content metadata
- **Machine Learning Model:**  
The system depends on the availability of a trained and validated machine learning model capable of generating accurate and relevant recommendations.
- **API Services:**  
The recommendation system depends on secure internal APIs for exchanging data with:
  - User Profile Service
  - Content Management Service
  - Viewing History Service
  - Front-end applications
- **Monitoring and Logging Services:**  
The recommendation system depends on monitoring and logging services to track system health, recommendation performance, operational metrics, and error events.

### 3. System Features:

#### a. Personalized Recommendations:

The system shall generate personalized recommendations based on the user's viewing history, preferences, and interaction data.

**b. Recommendation Ranking:**

The system shall rank recommended content based on relevance and display the Top-N recommendations.

**c. Metadata Processing:**

The system shall process and utilize content metadata (e.g., genre, cast, director, keywords, description) to generate recommendations.

**d. Similar User Preference Analysis:**

Recommendations should be generated based on similar users. Hence, following collaborative recommendations.

**e. Recommendation Refresh:**

The system shall automatically update recommendations when user preferences or interaction history change.

**f. Error Handling:**

The system shall notify the requesting application when recommendations cannot be generated due to missing or invalid input data.

**g. Logging:**

The system shall log recommendation requests, system errors, and operational events for monitoring and troubleshooting purposes.

**h. Recommendation History:**

The system shall maintain logs of generated recommendations for performance analysis and auditing.

**4. External Interface Requirements:**

**a. User Interfaces:**

No external interface required by the model. The existing interface just needs to call requests for recommended titles using the model via an API.

**b. Hardware Interfaces:**

**i. Application Server:**

The recommendation engine shall execute on cloud-based application servers capable of handling recommendation requests.

**ii. Database Server:**

The system shall interface with database servers storing movie metadata and user interaction data.

**iii. Storage Infrastructure:**

The system shall interface with cloud storage for storing datasets, trained recommendation models, and recommendation logs.

**c. Software Interfaces:**

**i. Recommendation API:**

The recommendation model shall expose a REST API that accepts recommendation requests and returns a ranked list of recommended movies and TV shows.

**ii. Content Metadata Database:**

The recommendation model shall utilize the movie metadata database to retrieve content attributes such as title, genre, cast, director, description, release year, language, and other metadata required for generating recommendations.

**iii. User Interaction Database:**



The recommendation model shall access user interaction data, including viewing history, ratings, watchlist information, and previously watched content, to generate personalized recommendations.

iv. **Logging and Monitoring Service:**

The recommendation model shall have an interface with centralized logging and monitoring services to record recommendation requests, response times, errors, and operational metrics for system monitoring and troubleshooting.

d. **Communications Interfaces:**

i. **RESTful API Communication:**

The recommendation model shall communicate with external applications through RESTful API endpoints.

ii. **HTTPS Protocol:**

All communication between client applications and the recommendation model shall be encrypted using HTTPS with Transport Layer Security (TLS).

iii. **JSON Data Format:**

Recommendation requests and responses shall be exchanged using the JSON data format.

iv. **Request-Response Communication:**

The recommendation model shall support synchronous request-response communication, where each recommendation request receives a corresponding response containing the recommended content.

v. **Error Communication:**

The recommendation model shall return standardized HTTP status codes and descriptive error messages for invalid requests, unavailable resources, or internal processing failures.

vi. **Authentication and Authorization:**

API requests to the recommendation model shall be authenticated and authorized to ensure that only approved applications and services can access recommendation functionality.

5. **Nonfunctional Requirements:**

a. **Performance Requirements:**

i. **Response Time:**

The system shall respond within 2 seconds under normal operating conditions.

ii. **Scalability:**

The system shall handle millions of recommendation requests simultaneously.

iii. **Availability:**

The recommendation service should be operational and accessible at least 99.9% of the time.

iv. **Throughput:**

The system shall process multiple concurrent recommendation requests while maintaining consistent response times.

- v. **Fault Recovery:**  
The system shall recover automatically from temporary infrastructure failures with minimal disruption to recommendation services.
  - vi. **Resource Utilization:**  
The system shall efficiently utilize computing resources to minimize operational costs while maintaining performance.
- b. **Safety Requirements:**
  - i. The system shall ensure that failures in the recommendation engine do not affect the availability of core Netflix streaming services.
  - ii. The system shall return a safe default response when recommendations cannot be generated.
  - iii. The system shall prevent corrupted or incomplete recommendation results from being displayed to users.
  - iv. The system shall maintain data integrity during failures or unexpected system interruptions.
  - v. System failures shall be logged for investigation and recovery.
- c. **Security Requirements:**
  - i. **Authentication:**  
The recommendation service shall only accept requests from authenticated users.
  - ii. **Authorization:**  
Only authorized personnel shall be permitted to modify recommendation models, datasets, or configuration settings.
  - iii. **Data Encryption:**  
Sensitive user data shall be encrypted during transmission and storage using industry-standard encryption mechanisms.
  - iv. **Privacy Protection:**  
The recommendation system shall process user data in accordance with applicable privacy regulations and organizational policies.
  - v. **Secure API Communication:**  
All communication between system components shall use secure communication protocols (HTTPS/TLS).
  - vi. **Audit Logging:**  
Administrative activities and system modifications shall be recorded in audit logs.
  - vii. **Data Access Control:**  
Access to user interaction data shall be restricted based on user roles and responsibilities.
  - viii. **Secure Storage:**  
Recommendation models, configuration files, and datasets shall be stored securely to prevent unauthorized access or modification.
- d. **Software Quality Attributes:**
  - i. **Reliability:**

The recommendation system shall consistently generate accurate and reliable recommendations during normal operation.

ii. **Availability:**

The system shall remain available to support recommendation requests with minimal downtime.

iii. **Scalability:**

The system shall support future growth in users, recommendation requests, and content volume without requiring major architectural changes.

iv. **Maintainability:**

The system shall be designed to support model updates, bug fixes, and feature enhancements with minimal impact on production services.

v. **Usability:**

The recommendation results shall be presented in a clear, intuitive, and user-friendly manner.

vi. **Performance Efficiency:**

The system shall generate recommendations with minimal latency while efficiently utilizing available computational resources.

vii. **Portability:**

The recommendation system shall be deployable across different cloud environments and computing platforms with minimal configuration changes.

viii. **Testability:**

The system shall support unit, integration, system, and performance testing to ensure software quality.